

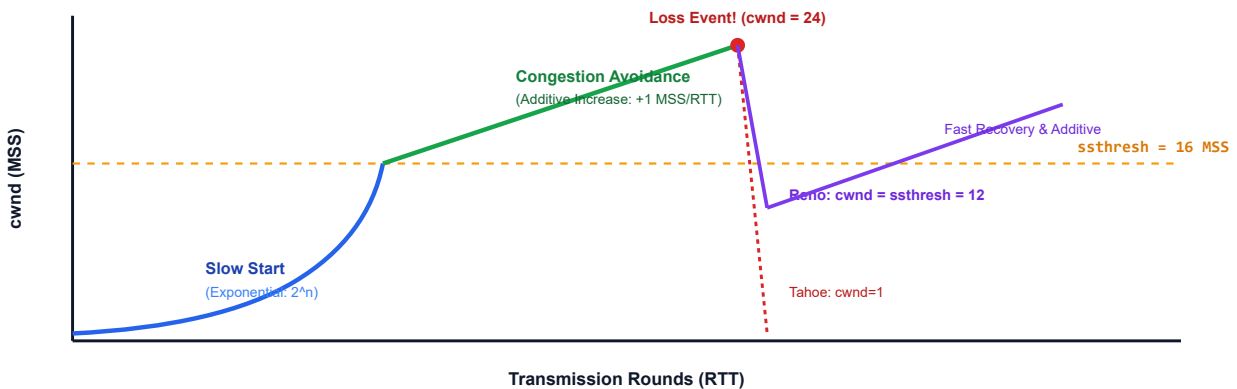
Module 08: TCP Congestion Control: Slow Start and Congestion Avoidance

Unit 4: Transport Layer

Congestion Dynamics & AIMD

Congestion Window (cwnd) vs Receiver Window (rwnd), Slow Start Exponential Ramp-Up, Congestion Avoidance Linear Growth, and AIMD Stability.

TCP Congestion Control: Slow Start, Congestion Avoidance & AIMD Dynamics



Mathematical Formulation of TCP Congestion Window (cwnd)

1. Slow Start & Additive Increase:

$cwnd = cwnd + 1 \text{ MSS per ACK}$
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1. Congestion Window Growth Rules

Slow Start: $cwnd = cwnd + 1 \text{ MSS per ACK}$ (Doubles every RTT) until $cwnd \geq ssthresh$.

Congestion Avoidance: $cwnd = cwnd + 1/cwnd \text{ per ACK}$ (Increases linearly by +1 MSS per RTT).

Multiplicative Decrease: Upon congestion, set $ssthresh = cwnd / 2$.

2. AIMD Sawtooth Dynamics

Additive Increase Multiplicative Decrease ensures stable network equilibrium by linearly probing available bandwidth and halving upon packet loss.